

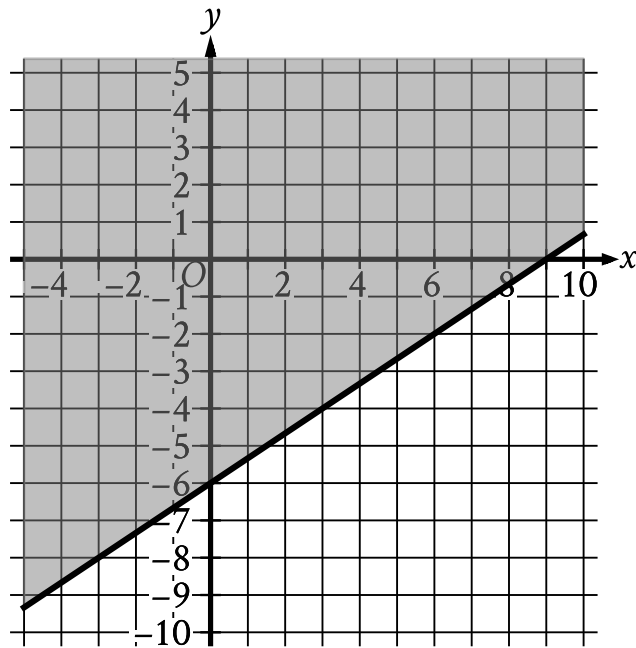
☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

07



- The boundary of the inequality is a solid line.
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (0 comma negative 6)
 - (9 comma 0)
- The area above and to the left of the boundary is shaded.

The shaded region shown represents the solutions to which inequality?

A. $y \geq \frac{2}{3}x - 6$

B. $y \geq \frac{2}{3}x + 6$

C. $y \geq \frac{2}{3}x - 9$

D. $y \geq \frac{2}{3}x + 9$

Q2

ALGEBRA

A chemist studying the impact of salt on a process mixes x kilograms of a low-salt mixture, which is 2% salt by weight, with y kilograms of a high-salt mixture, which is 96% salt by weight, to create 24 kilograms of a mixture that is 4% salt by weight. Which equation represents this situation?

- A. $0.96x + 0.02y = 0.0424$
- B. $0.02x + 0.96y = 0.0424$
- C. $0.96x + 0.02y = 24$
- D. $0.02x + 0.96y = 24$

Q3

ALGEBRA

A principal used a total of 25 flags that were either blue or yellow for field day. The principal used 20 blue flags. How many yellow flags were used?

- A. 5
- B. 20
- C. 25
- D. 30

The function f is defined by $f(x) = 8x$. For what value of x does $f(x) = 72$?

- A. 8
- B. 9
- C. 64
- D. 80

Hiro and Sofia purchased shirts and pants from a store. The price of each shirt purchased was the same and the price of each pair of pants purchased was the same. Hiro purchased 4 shirts and 2 pairs of pants for \$86, and Sofia purchased 3 shirts and 5 pairs of pants for \$166. Which of the following systems of linear equations represents the situation, if x represents the price, in dollars, of each shirt and y represents the price, in dollars, of each pair of pants?

A. $4x + 2y = 86$
 $3x + 5y = 166$

B. $4x + 3y = 86$
 $2x + 5y = 166$

C. $4x + 2y = 166$
 $3x + 5y = 86$

D. $4x + 3y = 166$
 $2x + 5y = 86$

Q6

ALGEBRA

Normal body temperature for an adult is between 97.8°F and 99°F , inclusive. If Kevin, an adult male, has a body temperature that is considered to be normal, which of the following could be his body temperature?

- A. 96.7°F
- B. 97.6°F
- C. 97.9°F
- D. 99.7°F

Q7

ALGEBRA

A total of 364 paper straws of equal length were used to construct two types of polygons: triangles and rectangles. The triangles and rectangles were constructed so that no two polygons had a common side. The equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. What is the best interpretation of $x, y = 24, 73$ in this context?

- A. If 24 triangles were constructed, then 73 rectangles were constructed.
- B. If 24 triangles were constructed, then 73 paper straws were used.
- C. If 73 triangles were constructed, then 24 rectangles were constructed.
- D. If 73 triangles were constructed, then 24 paper straws were used.

Q8

GRID-IN ALGEBRA

$$3x + 21 = 3x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions.
What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

The function f is defined by $f(x) = 25x + 30$. What is the value of $f(x)$ when $x = 2$?

- A. 50
- B. 57
- C. 80
- D. 110

Q10

ALGEBRA

$$\begin{aligned}2x + 7y &= 9 \\ 8x + 28y &= a\end{aligned}$$

In the given system of equations, a is a constant. If the system has infinitely many solutions, what is the value of a ?

- A. 4
- B. 9
- C. 36
- D. 54